Basketball Prospect Predictor

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Problem Statement

- **Question:** Can we predict NBA career success based on NCAA performance and qualitative analysis?
- Two signal sources
  - NCAA statistics - per-game box score numbers (points, rebounds, shooting efficiency, etc...)
  - Scouting reports - written text evaluations of each prospect's skills, athleticism, and projection
- Why a multimodal investigation?
  - Statistics miss character, athleticism and things that may not reflect on the scoreboard
  - Scouting reports are subjective & qualitative and it makes it hard to compare across scouts
  - Using them together should cancel their blind spots
  - Goal: train classifier/regression models on NCAA stats, along with a NLP approach to analyze scouting reports, and see better performance through a multi-modal approach than though any individual models

Data Collection

- Sources:
    - NCAA stats
        - Scraped from NCAA website
        - Backfill using sports dataverse
    - NBA stats
        - NBA API for basic metrics
        - Basketball Reference for VORP
        - databallr for DARKO
    - Scouting Reports from nbadraft.net

What Does the Data Look Like?

- As stated we have three sources of data, NCAA college stats, NBA career stats, and scouting reports
    - NCAA College stats
        - 787 players, draft classes 2009-2023
        - Stats: PPG, RPG, APG, FG%, 3P% FT% BPG, SPG, MPG, OReb, DReb, TO, Games, Height, Position, Class
    - NBA Career Stats (Best Season of First Three Years in NBA)
        - 787 matched players (same players from NCAA set), draft classes 2009-2023
        - Stats: PTS, REB, AST, STL, BLK, FG%, 3P%, FT%, +/-, MIN, GP
    - Scouting Reports
        - 1795 players, Draft classes 2009-2026
        - Reports from scouts that detail the strengths, weaknesses, NBA comparison, and a full narrative report about the player
- All three sources are joined on player name & draft year to form the full multi-modal training dataset

Regression

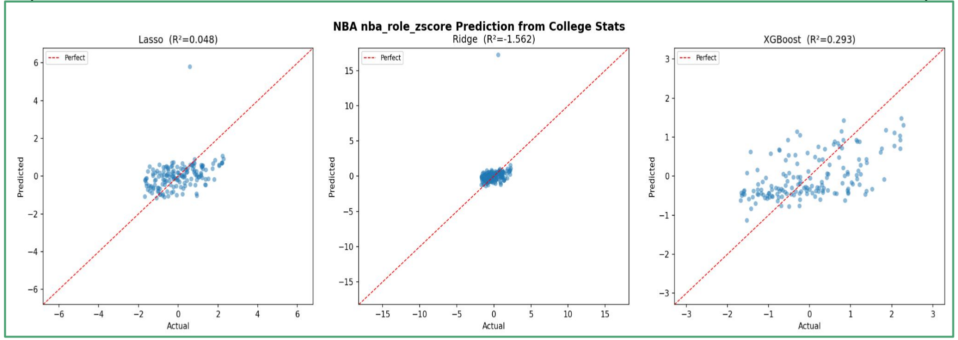
- Model Selection
  - NCAA stats produce a continuous outcome (role z-score) so regression seemed like the natural fit
  - We chose three variants:
    - **Lasso**
      - We have about 18 stats and they are highly correlated, the L1 penalty automatically zeros out redundant features (built-in feature selection)
    - **Ridge**
      - Softer comparison that shrinks weights rather than zeroing them (better when features contribute a little)
    - **XGBoost**
      - Captures non-linear interactions linear models miss (ex. High efficiency & high assist rate together predicting succeeds better than either alone)

- Model Implementation
- **Input**
  - NCAA stats ➔ Imputer ➔ StandardScaler ➔ $x \in \mathbb{R}^d$
  - Position ➔ OneHotEncoder([Guard, Forward, Center])
  - Class yr ➔ OrdinalEncoder ➔ StandardScaler
  - Stats: pts_pg, reb_pg, ast_pg, blk_pg, stl_pg, fg_pct, ft_pct, fg3_share, ft_rate, pts_per_fga, height_dev, team_difficulty
- **Target — nba_role_zscore**
  - z = Σ w□ · clip(z□, ±2.5) / Σ 1[s available]
  - Weights: MIN=0.30, pts=0.25, reb=0.20,
  - ast=0.15, stl=0.05, blk=0.05
  - No NBA data ➔ floor of −3.0
- **Lasso**
  - $L = \|y - Xw\|_2^2 + \alpha\|w\|_1$
  - α: 100 log-spaced values [1e-4, 100], 5-fold CV
  - Retained only 3 features: RPG, BKPG, FG%
- **Ridge**
  - $L = \|y - Xw\|_2^2 + \alpha\|w\|_2^2$
  - Same α search — keeps all features, shrinks weights
- **XGBoost**
  - ŷ = Σ□ f□(x)
  - GridSearchCV: depth [2,3], lr [0.05,0.1],
  - n_estimators [100,200], L1/L2 regularization
- **Output**
  - $\hat{y} \in \mathbb{R}$ (predicted nba_role_zscore)
  - > 1.5 = Star | 0.5–1.5 = Starter
  - −0.5–0.5 = Bench | < −0.5 = Bust

Regression

- Model Performance

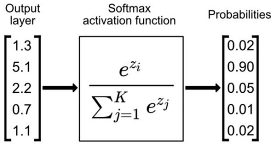
Model	R^2	RMSE	MAE
Lasso	0.0479	0.9811	0.7608
Ridge	-1.562	1.609	0.8405
XGBoost	0.2934	0.8452	0.7035



- Limitations:
  - NBA Outcomes depend on team fit, injuries, and player development which aren't in the NCAA data
  - Raw counting stats are position cofounded (center vs guard RPG mean very different things)
  - Noisy continuous target (binned classification is more tractable for this dataset)
- Future Directions
  - Position relative input normalization (z-score each stat within position group)
  - Draft class normalization to control for era effects
  - Feed continuous score into multimodal meta model alongside classification probability outputs

Classification

- Why classification?
- Tiered by distributions of key stats
  - Bust: z < -0.5
  - Bench: -0.5 <= z < 0.5
  - Starter 0.5 <= z < 1.5
  - Star z >= 1.5
- Logistic Regression (L1/L2)
  - Extension of linear models to classification
  - Softmax ➜



- XGBoost
  - Tier boundaries not linearly seperable
  - Tree structure refines decision making process by capturing combinations of features

- Input
  - NCAA stats ➜ median ➜ StandardScaler ➜ $x \in \mathbb{R}^d$
  - Position ➜ OneHotEncoder([G, F, C]) (optional)
  - Class Yr ➜ OrdinalEncoder ➜ Standard Scaler

Logistic Regression — L1 & L2

$$\hat{p}(y = k \mid x) = \frac{\exp(w_k^\top x + b_k)}{\sum_j \exp(w_j^\top x + b_j)}$$

$$\mathcal{L} = -\sum_i \log \hat{p}(y_i \mid x_i) + \frac{1}{C}\|w\|_p$$

- $C \in \mathrm{logspace}(-3,\ 2,\ 20)$, selected by 5-fold CV; metric: F_1-macro
- L1: solver $= \mathrm{SAGA}$ L2: solver $= \mathrm{LBFGS}$

XGBoost

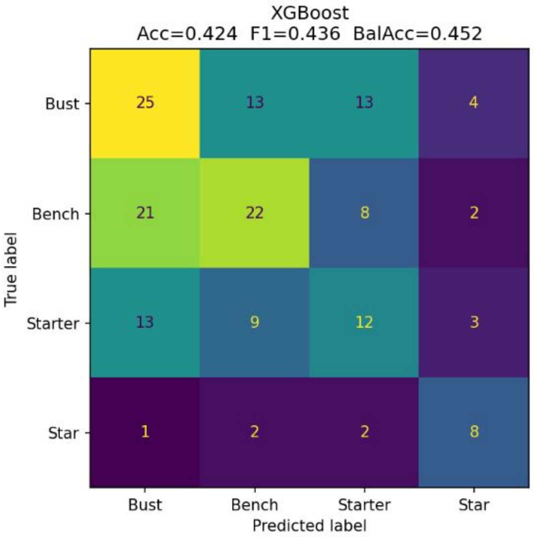
$$\hat{p} = \mathrm{softmax}\left(\sum_t f_t(x)\right), \quad \text{objective: multi:softprob}$$

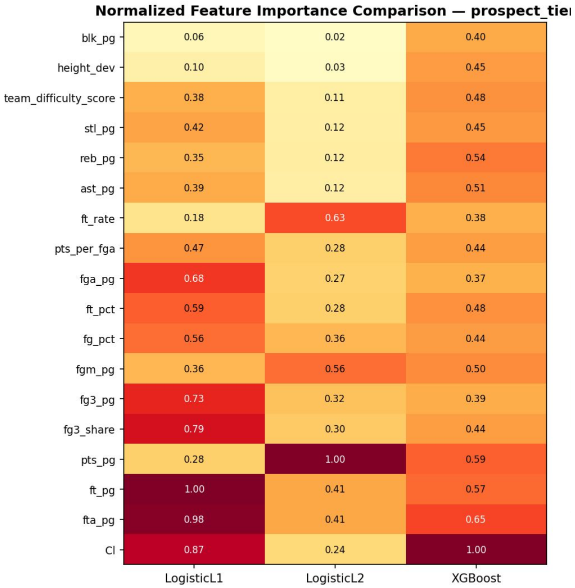
- GridSearchCV: $d \in \{2, 3\}$, $\eta \in \{0.05, 0.1\}$, $T \in \{100, 200\}$, $s \in \{0.7, 0.9\}$

- Class imbalance – essential
  - w_i = N / (K × N□)
  - N = total, K = num classes, N□ = class count
- Output
  - p̂ = [p_bust, p_bench, p_starter, p_star] ∈ [0,1]

Classification

Model	Accuracy	F1-macro	Bal.Acc
LogisticL1	0.3228	0.3154	0.3528
LogisticL2	0.3228	0.3160	0.3547
XGBoost	0.4241	0.4365	0.4523





- Limitations
  - Personality, work ethic, injuries, etc
  - Arbitrary boundary decisions and player classification (what makes a player 'good')
    - Impact vs presence
  - Star class underrepresented
  - Ordinal structure ignored
    - Adjacent errors should be penalized less than distant ones
- Future Directions
  - Ordinal regression so we can respect rank ordering of the tiers
  - Probability Calibration
    - Add isotonic regression so probability outputs are reliable for meta model input
  - Independent study - what makes an NBA player 'good'

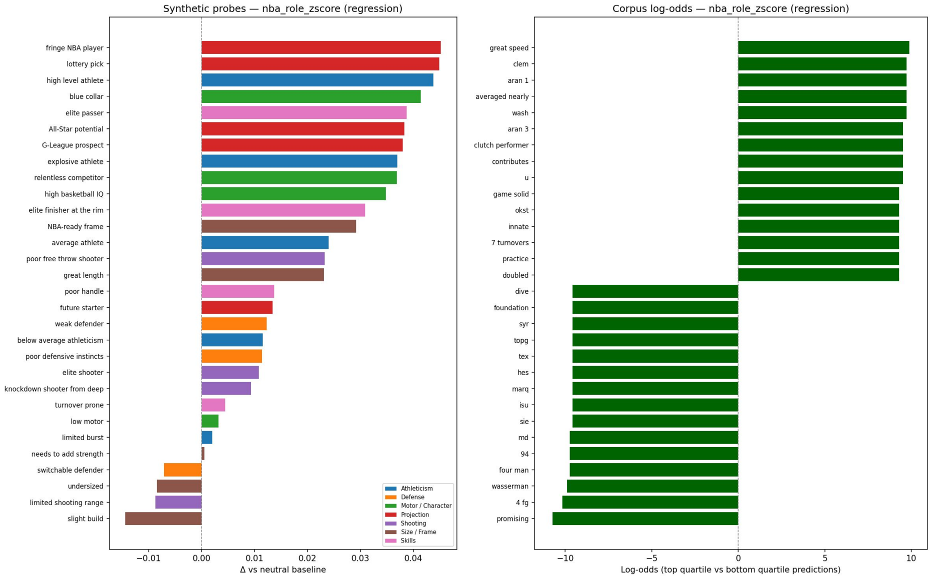
Text – Version 1

- Model Selection
  - Uses DistilBERT on scouting reports
    - Short but context-heavy text
    - Captures nuanced language beyond box score
  - Report → compact deep embedding
  - Can handle regression + classification
    - Z score, VORP, DARKO
    - Prospect tier
      - Derived from Z score

- Model Implementation
  - Tokenized and encoded with DistilBERT
    - Batch size of 32
  - Pooled text representation into:
    - MLP block: Linear(768 → 64) → ReLU → Dropout
    - Shared hidden layer: Linear(64 → 32) → ReLU → Dropout
  - Task-specific output head:
    - Regression output: one continuous value per player
      - Linear(32 → 1)
    - Classification output: 4 class logits (converted to probabilities)
      - Linear (32 → 4)

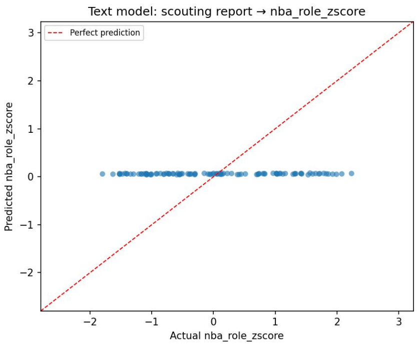
Text – Version 1

- Interpretation Approach
  - Gut check… what does a report for a good NBA player look like
  - Use chosen phrases to test how specific scouting language shifts model output
  - Compare language patterns in high-prediction vs low-prediction reports
  - Analyze tone across strengths, weaknesses, and outlook sections against predictions



Text – Version 1

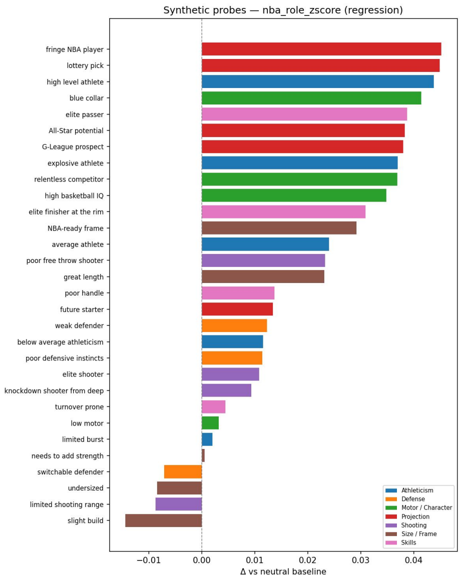
- Model Performance
  - Results were not good
    - Regression: MAE ≈ 0.92
    - Classification: Accuracy ≈ 0.29, Macro-F1 ≈ 0.17
  - Model captures *some* signal,
    - Not strong enough as a standalone predictor

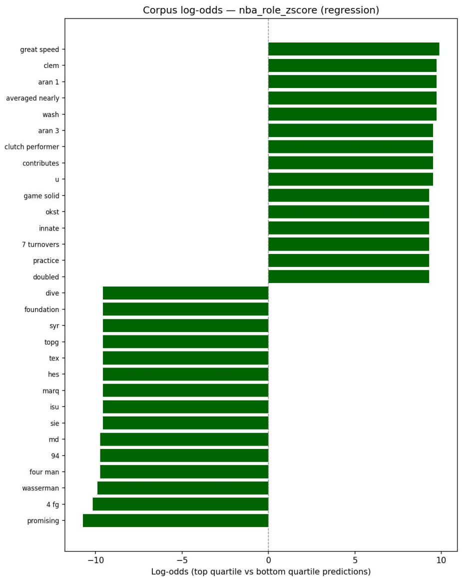


- Limitations + Future Direction
  - Scouting language is noisy and inconsistent across sources
    - A lot of irrelevant information
  - Text coverage and detail vary by player, which weakens supervision quality
  - Use interpretability outputs to identify which phrasing patterns are actually predictive
  - Primary role going forward: complementary feature source, not standalone model

Text – Version 2

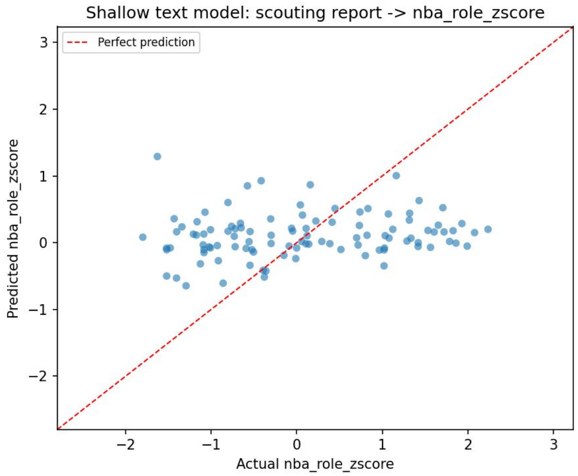
- Model Selection
  - The idea: maybe simpler model will perform better
    - Less prone to noise distraction?
  - Shallow model focused on speed and transparency
  - Uses sentiment and scouting-term signals
- Model Implementation
  - Extract VADER sentiment
  - Count set of good/bad words/phrases
  - Train Ridge regression model
  - Convert into tier probabilitirs





Text – Version 2

- Model Performance
  - Results were not good
    - MAE ≈ 0.90, R^2 ≈ 0.00
  - Again, there is *some* signal
    - But not very strong



- Limitations + Future Direction
  - Handcrafted language features can miss context and subtle phrasing
  - Feature quality depends on lexicon coverage and report consistency
  - Goal is not text-only dominance, but better complementary signal for final stacking

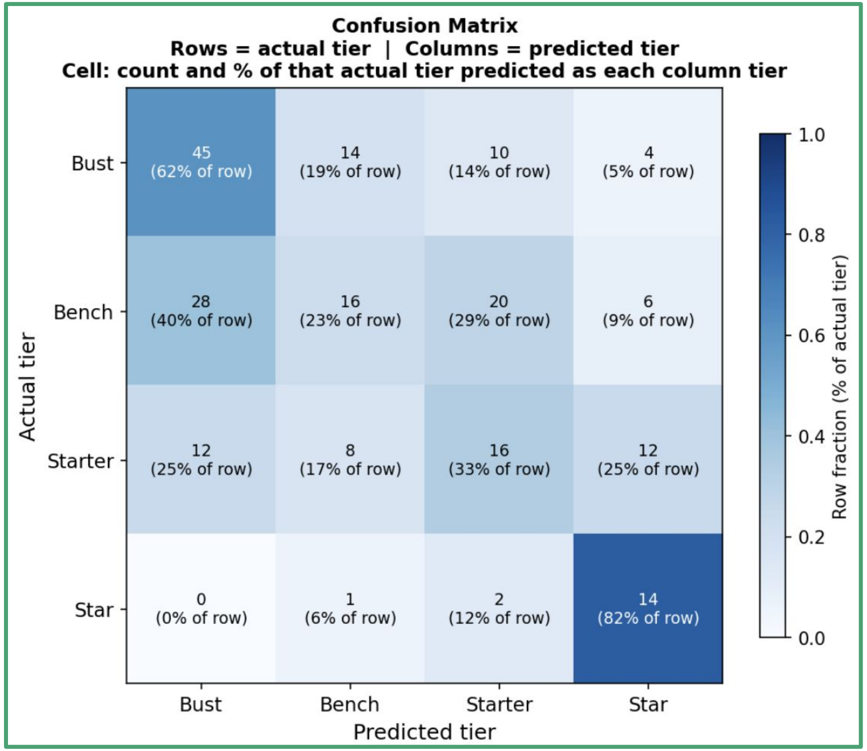
```
[[log_reg: p_bust, p_bench…],
[xgb_reg: p_bust, p_bench…],
[log_cl: p_bust, p_bench…],
[xgb_cl: p_bust, p_bench…],
[nlp: p_bust, p_bench…]]
```

Multi-Modal Meta-Model

- Why MMMM
  - Classification optimizes for boundaries
  - Regression captures **where** in bound a player is expected to fall
  - Text (in theory) captures non-statistical factors (attitude, work ethic, etc)
- Unified Probability Interface
  - Base models output in the same format
  - Regression z-scored converted to probabilities using Gaussian CDF
  - Enables modular base model swapping via config

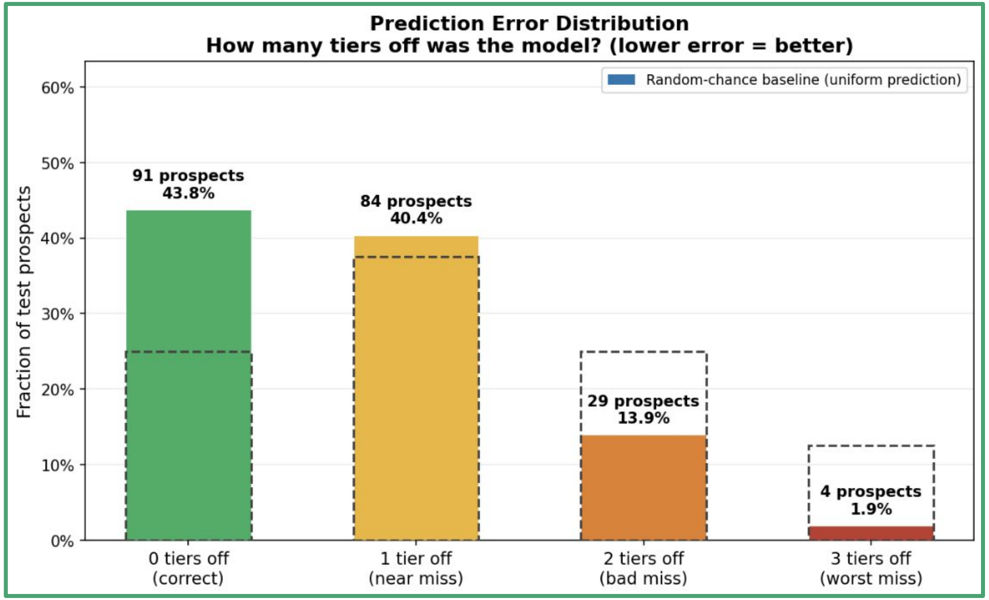
- Stacking Architecture
  - Base model p(k) concatenated into a meta-feature vector
  - Logistic regression (meta-model) learns which base models to trust for which tiers
  - Weights = reliability of each model
- Out-of-Fold (OOF) Training
  - Base model trained on k-1 folds and predicts held out fold
  - MMMM trained only on OOF predictions
  - Prevents MMMM from seeing same training data 4 times

Meta-Model Results

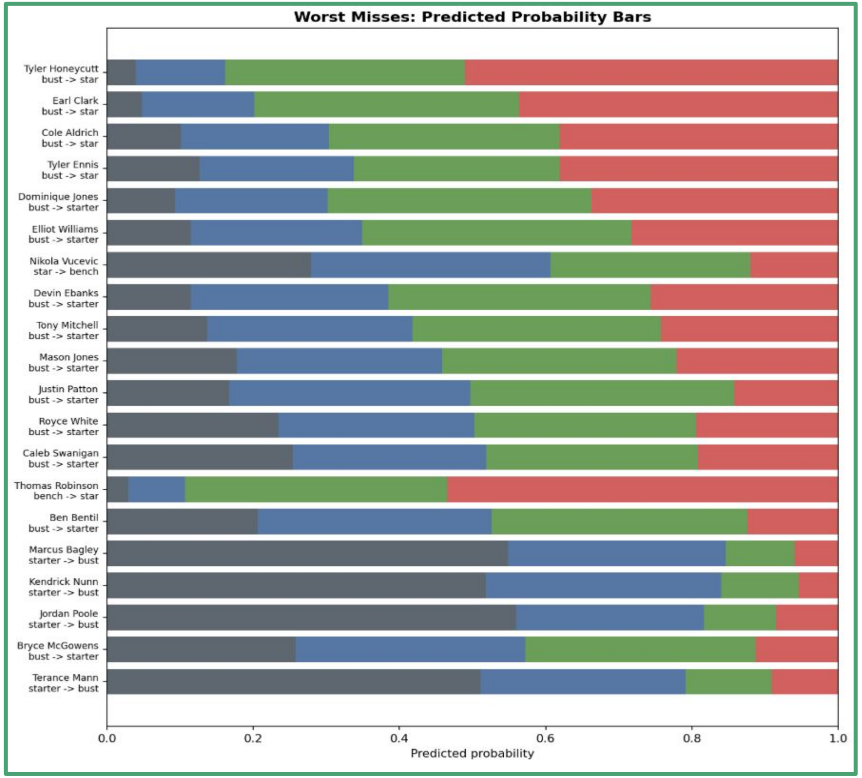


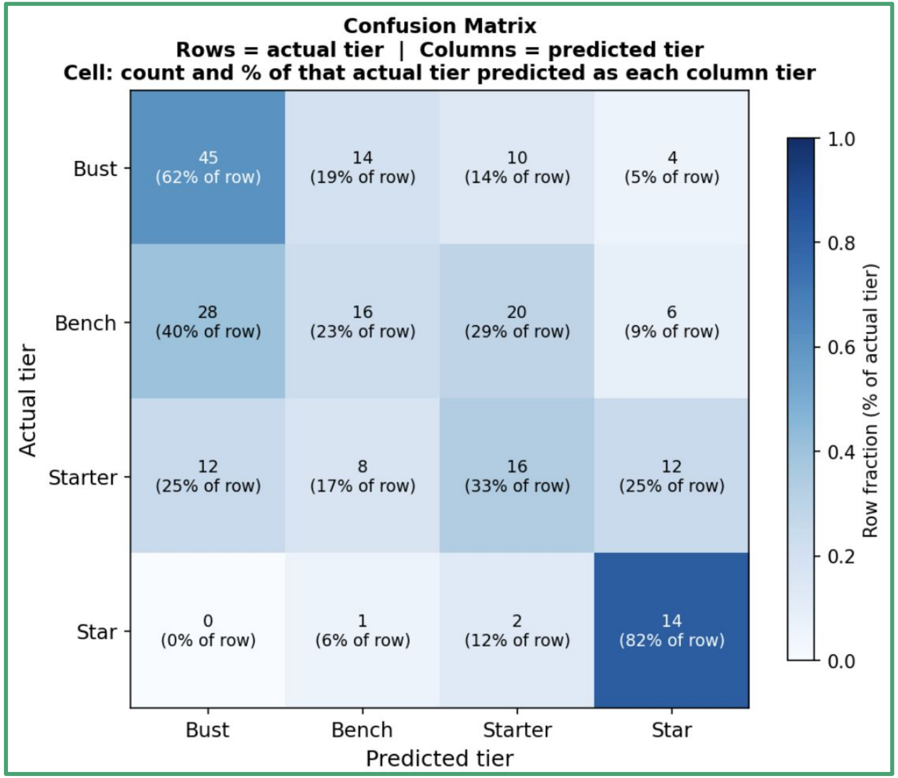
Multimodal Model Comparison
(blue row = final stacker output; other rows = individual base model performance)

Model	Acc	Within 1	Dist Wt Acc	Ord MAE	Exp MAE	QWK	F1 Macro
Stacker (final)	**0.4375**	**0.8413**	**0.7532**	**0.7404**	**0.7191**	**0.4902**	**0.4312**
CLF - L2 Logistic	0.3558	0.7548	0.6987	0.9038	0.7476	0.1818	0.3442
CLF - XGBoost	0.4135	0.7885	0.7276	0.8173	0.7453	0.1996	0.3285
REG - Ridge	0.3846	0.9279	0.7708	0.6875	0.7141	0.3801	0.2827
REG - XGBoost	0.3846	0.9519	0.7788	0.6635	0.6728	0.4544	0.3047
text__scouting	0.3798	0.9087	0.7628	0.7115	0.8425	0.1807	0.2166
text__scouting_lexico	0.3990	0.8654	0.7516	0.7452	0.8399	0.2203	0.2631



Meta-Model Results





Future Direction

1. Improving data processing approach
2. Getting combine data as a proxy for physicality & health
3. Implementing proprietary classification/regression models to respect ordinality
4. Further exploring NLP and different scouting sources
5. Hyperparameter optimization
6. Conflict between predicting a continuous output and placing into buckets

Predicting the 2026 NBA Draft